

LEWISTON, ID, USA

WMO: 727830

Lat: 46.375N

Lon: 117.016W

Elev: 1436

StdP: 13.95

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24149

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.8	19.4	3.9	7.1	16.9	10.3	9.8	21.3	27.7	42.3	24.2	42.2	3.9	70	0.498

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.3	98.7	65.3	95.4	64.4	91.8	63.3	67.8	92.0	66.1	89.8	64.7	87.3	6.6	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	82.0	72.4	58.0	75.8	71.6	56.0	70.5	71.0	32.7	91.7	31.4	90.0	30.4	87.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.8	17.7	14.8	8.0	105.1	6.8	3.0	3.1	107.3	-0.8	109.1	-4.6	110.8	-9.5	113.0
			7.0	70.9	6.8	2.2	2.1	72.5	-1.8	73.8	-5.6	75.0	-10.5	76.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.7	36.2	39.0	45.1	51.3	59.7	66.4	76.3	75.2	65.8	52.3	41.7	34.8	(6)
	DBStd	15.85	7.89	7.26	6.20	6.18	7.19	6.97	6.16	6.10	6.98	6.91	7.36	7.97	(7)
	HDD50	1760	428	310	173	54	7	0	0	0	1	52	261	474	(8)
	HDD65	5024	892	728	617	412	197	63	3	5	73	398	699	937	(9)
	CDD50	3123	1	2	22	93	307	492	815	781	475	121	12	2	(10)
	CDD65	913	0	0	0	2	32	105	353	320	98	3	0	0	(11)
	CDH74	11545	0	0	0	46	447	1331	4608	3925	1148	40	0	0	(12)
	CDH80	5717	0	0	0	7	137	539	2521	2059	448	6	0	0	(13)
(14) Wind	WSAvg	5.4	5.9	5.6	5.9	6.0	5.4	5.5	5.4	5.1	4.6	4.6	5.2	5.7	(14)
(15) Precipitation	PrecAvg	12.1	1.2	0.8	1.1	1.2	1.5	1.2	0.6	0.7	0.7	0.9	1.1	1.1	(15)
	PrecMax	17.6	3.1	2.2	2.4	3.3	3.8	2.7	2.6	3.0	2.5	2.4	2.7	2.9	(16)
	PrecMin	7.5	0.2	0.2	0.2	0.1	0.2	0.2	0.0	0.0	0.0	0.0	0.2	0.1	(17)
	PrecStd	2.2	0.7	0.5	0.5	0.7	0.9	0.6	0.6	0.7	0.6	0.6	0.6	0.6	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.7	58.6	68.8	79.9	89.4	97.8	103.3	102.3	95.6	79.4	63.8	56.8	(19)
		MCWB	46.6	48.3	52.4	58.7	63.2	66.5	67.2	65.3	64.4	59.4	54.0	47.8	(20)
	2%	DB	51.4	54.4	62.9	73.3	83.9	90.8	99.6	98.2	89.7	72.6	58.3	50.9	(21)
		MCWB	44.2	45.4	49.7	56.1	60.7	64.2	66.0	64.6	61.8	55.5	49.7	44.1	(22)
	5%	DB	48.8	51.5	59.3	68.3	78.9	86.1	96.4	94.8	85.1	67.4	54.9	47.6	(23)
		MCWB	42.4	43.9	48.1	53.6	59.0	62.4	64.8	63.7	60.2	53.2	47.8	41.9	(24)
	10%	DB	46.3	48.9	55.8	63.3	74.3	81.7	92.8	91.1	80.8	63.3	51.7	44.8	(25)
		MCWB	40.8	42.3	46.2	50.6	57.0	60.4	63.8	62.6	58.8	51.3	45.6	40.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.2	49.9	53.9	60.7	65.3	68.6	71.2	68.2	66.7	60.2	55.6	49.2	(27)
		MCDB	54.0	57.0	65.2	76.5	83.6	92.8	94.4	95.5	88.5	75.5	63.1	56.3	(28)
	2%	WB	44.9	47.0	51.3	57.1	62.6	66.0	68.4	66.4	63.8	57.2	51.3	45.1	(29)
		MCDB	50.6	52.6	60.1	71.1	79.5	87.4	92.2	92.2	83.4	68.8	56.7	50.1	(30)
	5%	WB	43.1	44.7	49.4	54.3	60.4	64.0	66.7	65.1	61.8	54.9	48.7	42.4	(31)
		MCDB	48.1	50.3	58.2	66.5	75.1	82.9	90.4	89.9	81.5	65.2	53.5	46.6	(32)
	10%	WB	41.3	42.8	47.2	51.7	58.5	61.9	65.2	63.8	59.8	52.7	46.4	40.6	(33)
		MCDB	45.7	48.1	55.0	61.7	71.8	78.5	88.8	87.2	78.4	61.6	51.1	44.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	13.7	17.5	19.6	21.9	23.1	28.3	27.9	25.2	18.9	12.7	9.8	(35)
		MCDBR	13.1	16.6	23.0	28.4	29.7	31.2	33.4	33.2	32.3	25.1	15.8	12.6	(36)
	5% WB	MCWBR	8.5	10.0	13.1	14.6	12.6	11.9	11.8	12.2	13.0	12.4	9.7	8.3	(37)
		MCWBR	12.5	15.4	20.9	26.0	26.3	28.3	29.9	29.7	28.8	22.0	14.3	11.7	(38)
(40) Clear-Sky Solar Irradiance	taub		0.295	0.299	0.318	0.340	0.358	0.358	0.359	0.367	0.347	0.329	0.300	0.293	(40)
		taud	2.471	2.505	2.488	2.420	2.393	2.412	2.413	2.384	2.418	2.472	2.509	2.458	(41)
	Ebn at Noon		257	281	290	291	288	287	285	279	275	264	253	244	(42)
		Edh at Noon	21	25	29	34	36	36	36	35	31	25	20	19	(43)
(44) All-Sky Solar Radiation	RadAvg		397	680	1033	1507	1881	2129	2386	1971	1442	886	461	311	(44)
	RadStd		34	76	102	131	150	154	106	127	114	79	39	33	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+175	+108
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	N/A

Nomenclature: See separate page